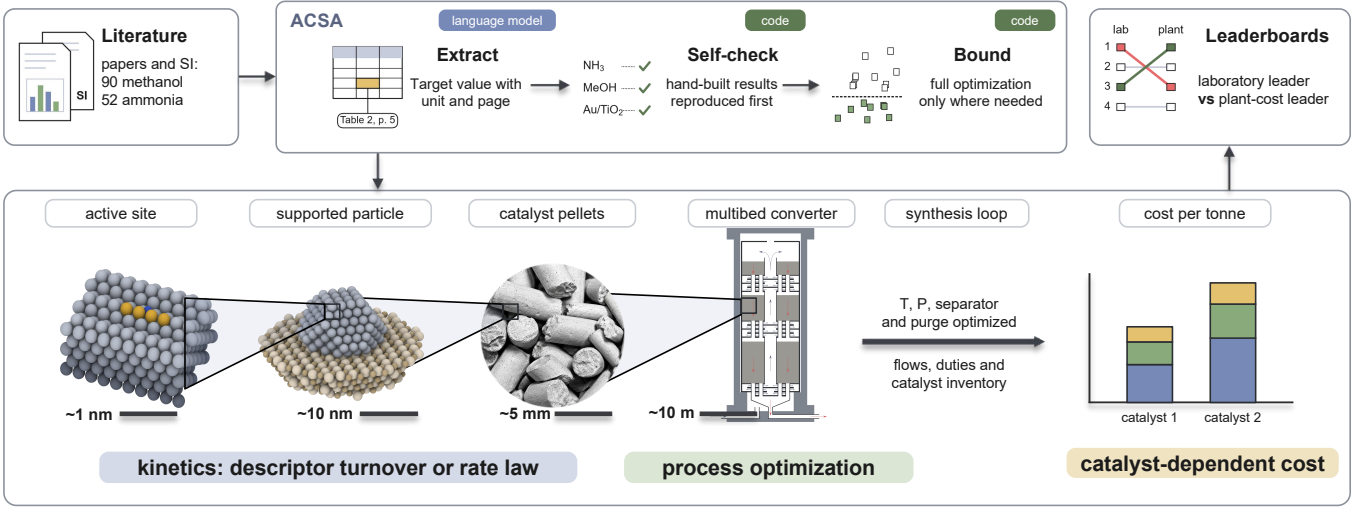

Literature
papers and SI:
90 methanol
52 ammonia
SI
ACSA
language model
code
code
Extract
Target value with unit and page
Table 2, p. 5
NH3
MeOH
Au/TiO2
Self-check
hand-built results reproduced first
Bound
full optimization only where needed
lab
plant
1
2
3
4
Leaderboards
laboratory leader
vs plant-cost leader
active site
supported particle
catalyst pellets
multibed converter
synthesis loop
cost per tonne
~1 nm
~10 nm
~5 mm
~10 m
T, P, separator and purge optimized
flows, duties and catalyst inventory
catalyst 1
catalyst 2
kinetics: descriptor turnover or rate law
process optimization
catalyst-dependent cost